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Development, Optimization and Validation of a Silver-Deposited Silicon SERS Sensor for Molecular Detection

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Abstract

Surface-Enhanced Raman Scattering (SERS) is a powerful analytical technique capable of detecting trace amounts of molecules through plasmonic enhancement generated by metallic nanostructures. In this study, a silver-deposited silicon-based SERS substrate was fabricated using an HF-assisted electroless deposition process and systematically optimized using Rhodamine 6G (R6G) as a model analyte. Two fabrication conditions were investigated using 5 mL HF and 10 mL HF, combined with four silver deposition soaking times (3, 4, 5, and 6 minutes). A total of 97 Raman spectra were collected using a 785 nm Raman system at 10% laser power, 5 s integration time, and 5 accumulations. A complete preprocessing workflow consisting of fingerprint extraction, baseline correction, cosmic ray removal, spectral smoothing, and vector normalization was implemented. Advanced statistical validation including peak detection, signal-to-noise ratio analysis, reproducibility assessment, correlation analysis, and PCA was performed. Results identified the 10HF–6 min condition as the best-performing substrate configuration. The developed platform demonstrates strong SERS enhancement and provides a foundation for future detection of biologically relevant molecules such as N-Acetylmuramic Acid (NAM).

1. Introduction

Surface-Enhanced Raman Scattering (SERS) enhances Raman signals through localized surface plasmon resonance generated by noble metal nanostructures. Silver is one of the most widely used plasmonic materials due to its strong enhancement capability in the visible and near-infrared spectral region.

The performance of a SERS substrate depends strongly on nanoparticle density, surface morphology, hotspot distribution, and fabrication reproducibility. Therefore, optimization of fabrication parameters is essential before applying the substrate to biological detection applications.

Rhodamine 6G (R6G) was selected as a standard Raman reporter molecule because its characteristic Raman peaks are well established and commonly used for evaluating SERS performance.

2. Objective

The objectives of this work were:

1. To fabricate silver-deposited silicon SERS substrates using HF-assisted electroless deposition.
2. To investigate the effect of HF concentration and soaking time on SERS enhancement.
3. To identify the optimal fabrication condition using R6G as a model analyte.
4. To evaluate substrate reproducibility, signal quality, and statistical reliability.
5. To establish a validated SERS platform for future detection of N-Acetylmuramic Acid (NAM).

3. Materials and Methods

3.1 Materials

Substrate Material

- Commercial silicon wafer

Chemicals

- Hydrofluoric Acid (HF)
- Silver Nitrate (AgNO_3)
- Acetone
- Deionized Water (DI Water)
- Rhodamine 6G (R6G)

Target Molecule Used in Validation

- Rhodamine 6G (1 mg/mL)

Future Target Molecule

- N-Acetylmuramic Acid (NAM)
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3.2 SERS Substrate Fabrication

The fabrication process consisted of the following steps:

Step 1: Cleaning

Silicon chips were cleaned using acetone followed by multiple rinses with deionized water.

Step 2: HF-Assisted Silver Deposition

Two fabrication solutions were prepared:

Substrate Type A

- 45 mL DI Water
- 5 mL HF
- 6 mg AgNO_3

Substrate Type B

- 45 mL DI Water
- 10 mL HF
- 6 mg AgNO_3

Step 3: Silver Deposition

Silicon chips were immersed in the prepared solutions for:

- 3 minutes
- 4 minutes
- 5 minutes

- 6 minutes

Step 4: Post-Treatment

After deposition:

- Chips were rinsed with DI water
- Dried using nitrogen gas
- Stored until Raman measurements

This resulted in eight experimental conditions:

Condition	HF Volume	Soaking Time
3min-5HF	5 mL	3 min
4min-5HF	5 mL	4 min
5min-5HF	5 mL	5 min
6min-5HF	5 mL	6 min
3min-10HF	10 mL	3 min
4min-10HF	10 mL	4 min
5min-10HF	10 mL	5 min
6min-10HF	10 mL	6 min

4. Raman Measurements

Raman spectra were acquired under identical experimental conditions.

Parameter	Value
Excitation Wavelength	785 nm
Laser Power	10%
Integration Time	5 s
Accumulations	5
Spectral Points	2048
Raman Range	-46 to 2842 cm ⁻¹

A total of 97 Raman spectra were collected.

5. Dataset Summary

The final dataset contained:

Category	Count
Total Spectra	97
5HF Spectra	49
10HF Spectra	48
3 min	25
4 min	22
5 min	25
6 min	25

Dataset auditing confirmed:

- No missing files
- No duplicate files
- No missing metadata
- Uniform spectral length (2048 points)

6. Data Processing Pipeline

A multi-stage preprocessing workflow was implemented.

6.1 Fingerprint Extraction

Spectra were cropped to:

500–2000 cm^{-1}

to isolate the Raman fingerprint region.

Figure 1. Fingerprint Region Extraction

Explanation:

This figure shows the original Raman spectrum and the selected fingerprint region used for all subsequent analyses. The cropped region contains the most informative molecular vibrations while excluding non-informative spectral regions.

6.2 Baseline Correction

Asymmetric Least Squares (ALS) baseline correction was applied.

Figure 2. Baseline Correction Example

Explanation:

The figure demonstrates removal of fluorescence background from a representative spectrum. The corrected spectrum exhibits a flat baseline while preserving Raman peak integrity.

6.3 Cosmic Ray Removal

Cosmic ray spikes were removed using a median-based detection algorithm.

Figure 3. Cosmic Ray Removal Example

Explanation:

The figure illustrates detection and correction of isolated spike artifacts generated by detector noise without affecting genuine Raman peaks.

6.4 Spectral Smoothing

Savitzky-Golay smoothing was applied.

Figure 4. Spectral Smoothing Example

Explanation:

The figure compares spectra before and after smoothing, demonstrating significant noise reduction while preserving peak position and intensity.

6.5 Vector Normalization

Vector normalization was selected after comparison with Min-Max and SNV normalization.

Figure 5. Normalization Method Comparison

Explanation:

This figure compares different normalization approaches and demonstrates why vector normalization was selected for downstream quantitative analysis.

7. R6G Peak Detection Analysis

The following characteristic R6G Raman peaks were analyzed:

- 612 cm^{-1}
- 773 cm^{-1}
- 1184 cm^{-1}
- 1312 cm^{-1}
- 1361 cm^{-1}
- 1507 cm^{-1}
- 1650 cm^{-1}

Peak detection was performed using:

- Prominence = 0.01
- Width = 2
- Distance = 5

Figure 6. Mean R6G Spectrum

Explanation:

The mean spectrum obtained from all measurements showing the characteristic Raman peaks of R6G and confirming successful SERS enhancement.

Figure 7. Peak Detection Example

Explanation:

Automatically detected peaks are marked and matched to literature peak positions, validating the peak detection algorithm.

8. Substrate Optimization

Performance metrics including peak intensity, peak area, SNR, reproducibility, and correlation were compared.

Figure 8. HF Concentration Comparison

Explanation:

Comparison of mean Raman peak intensities between 5HF and 10HF substrates. The 10HF substrates consistently produced stronger Raman enhancement.

Figure 9. Soaking Time Optimization

Explanation:

Comparison of Raman intensity across soaking times. Signal intensity increased with soaking duration, reaching maximum enhancement at 6 minutes.

Figure 10. Final Substrate Ranking

Explanation:

Composite ranking of all eight fabrication conditions based on multiple performance metrics. The 6min-10HF condition achieved the highest overall score.

9. Advanced Statistical Validation

9.1 Peak Position Stability

All characteristic peaks exhibited standard deviations below 3 cm⁻¹.

Figure 11. Peak Position Stability

Explanation:
Shows Raman shift variability of all characteristic peaks. Low variability confirms excellent spectral stability.

9.2 Signal-to-Noise Ratio Analysis

Figure 12. SNR Heatmap

Explanation:
Displays average signal-to-noise ratios for all fabrication conditions. Higher values indicate stronger analytical performance.

9.3 Reproducibility Assessment

Figure 13. Peak Area Reproducibility

Explanation:
Comparison of coefficient of variation (CV%) values across characteristic Raman peaks. Lower CV values indicate higher reproducibility.

9.4 Correlation Analysis

Figure 14. Replicate Correlation Heatmaps

Explanation:
Pearson correlation matrices showing similarity among replicate measurements for each fabrication condition.

9.5 Principal Component Analysis

Figure 15. PCA Score Plot

Explanation:
Principal Component Analysis demonstrates clustering trends associated with fabrication conditions, confirming measurable differences between substrate types.

10. Key Results

The major findings are summarized below:

Metric	Result
Best HF Condition	10HF
Best Soaking Time	6 min
Best Overall Condition	6min-10HF
Performance Score	0.9484
Highest SNR Peak	1507 cm ⁻¹
Most Reproducible Peak	1184 cm ⁻¹
Mean Correlation	0.8757

11. Study Limitations

The present work focused on optimization using Rhodamine 6G.

The following analyses remain future work:

- SEM characterization
 - AFM characterization
 - Enhancement Factor (EF) calculation
 - Batch-to-batch reproducibility assessment
 - Direct NAM validation
 - Detection limit studies
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12. Conclusion

A silver-deposited silicon SERS substrate was successfully fabricated and optimized using HF-assisted electroless deposition. A total of 97 Raman spectra were collected and analyzed using a comprehensive preprocessing and validation workflow.

Among the investigated conditions, the 10HF substrate with 6 minutes soaking produced the strongest Raman enhancement and highest overall performance score. Statistical validation demonstrated excellent peak stability, strong signal-to-noise ratios, and good reproducibility.

The optimized substrate provides a reliable platform for future molecular detection studies and will be utilized for subsequent detection and validation of N-Acetylmuramic Acid (NAM).